

DEPARTMENT OF MECHANICAL ENGINEERING

SEMINAR SERIES

Intelligent Transportation Infrastructure for Autonomous Mobility: From Multimodal Sensing to Physical AI

Yiqiao Li, Ph.D.
Department of Civil Engineering
The City College of New York



Abstract

Advances in autonomous mobility require intelligent systems capable of perceiving, understanding, and predicting complex interactions in dynamic traffic environments. However, conventional vehicle-centric approaches remain limited by sensing constraints and incomplete awareness of system-level traffic dynamics. AI-enabled transportation infrastructure offers a promising approach to enhancing situational awareness, cooperation, and safety. This seminar presents a research framework for transforming transportation infrastructure from passive monitoring systems into intelligent cyber-physical systems. First, infrastructure-based sensing and 3D perception will be discussed, including LiDAR point-cloud reconstruction, multimodal sensor fusion, and learning-based vehicle characterization. Second, the seminar explores how vision-language models and domain-informed AI can bridge physical measurements and semantic understanding of complex transportation environments. Third, recent advances in roadside LiDAR-based cooperative perception and safety auditing will illustrate how infrastructure intelligence can improve environmental observability, characterize road-user interactions, and support safety assessment. Finally, the seminar will highlight emerging opportunities in Physical AI, including occupancy-based world models, behavior-aware prediction, and physically grounded traffic representations. These developments point toward intelligent infrastructure capable of anticipating traffic dynamics and supporting safety-critical decisions in connected and autonomous mobility.

Speaker Bio

Dr. Yiqiao Li is an Assistant Professor of Civil Engineering at The City College of New York (CCNY) and a faculty member in the Computer Science Program at the CUNY Graduate Center. She directs the AI & Mobility Research Lab at CCNY. Her research focuses on cyber-physical transportation systems, AI-enabled intelligent infrastructure, and cooperative driving, with applications in traffic safety, operations, and freight mobility. Her work integrates roadside LiDAR, computer vision, machine learning, and infrastructure-based perception to develop intelligent transportation systems that better understand, predict, and respond to dynamic traffic conditions. Dr. Li's recent research has been supported by the U.S. Department of Transportation (USDOT), the National Science Foundation (NSF), the New York City Department of Environmental Protection (NYC DEP), and other public sponsors.

Date: Thursday, October 15, 2026
Location: Steinman 254

Time: 2:00 PM
Host: Bo Wang

Attendance open to all interested persons

